Chapter 9

Externalities and Public Goods

ECON 003
UC Riverside · Summer 2026



Chapter 9 in Five Moves

One idea, five times over: PRIVATE cost and benefit are not SOCIAL cost and benefit.



What an externality is, and what it costs



Whether people can fix it themselves



What government can do about it



Goods the market will not supply



Goods the market uses up

Sections 9.1 → 9.5: EXTERNALITIES · PRIVATE SOLUTIONS · GOVERNMENT SOLUTIONS · PUBLIC GOODS · COMMON POOL RESOURCES

MARKET FAILURE — an inefficient allocation of resources. All five sections of this chapter are cases of it.

PART 1

When the Price Stops Telling the Truth

Externality - Outside Consequences

An EXTERNALITY occurs when an economic activity has either a spillover COST to, or a spillover BENEFIT for, a bystander.

INTERNAL COST

What the people in the market actually pay. Coal, turbines, wages, the loan. This is the firm's supply curve.



EXTERNAL COST

What the activity costs everybody else. Sulphur dioxide, mercury, a warmer planet. Nobody in the market is billed for it.



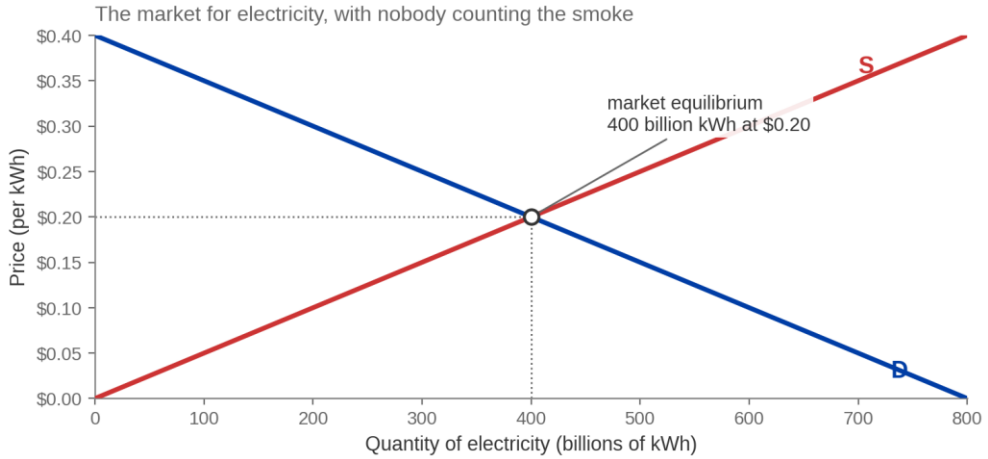
SOCIAL COST

The two added together. What the activity really costs the world. This is the curve the market never sees.

MARGINAL EXTERNAL COST — social cost minus private cost, per unit.

The Market for Electricity, As Chapter 4 Would Draw It

MARKET EQUILIBRIUM — 400 billion kWh at $0.20, where supply is MARGINAL PRIVATE COST only.



The supply curve is MARGINAL PRIVATE COST — what the plant pays. It is not MARGINAL SOCIAL COST.

Redrawn from Exhibit 9.1.

What That Diagram Does Not Show



Where does the smoke go on this graph?

Nowhere. Burning coal to make electricity releases sulphur dioxide and nitrogen oxides — lung irritation, bronchitis, pneumonia. At high doses the mercury from coal plants has been linked to birth defects. Emissions from power generation are part of the warming record.

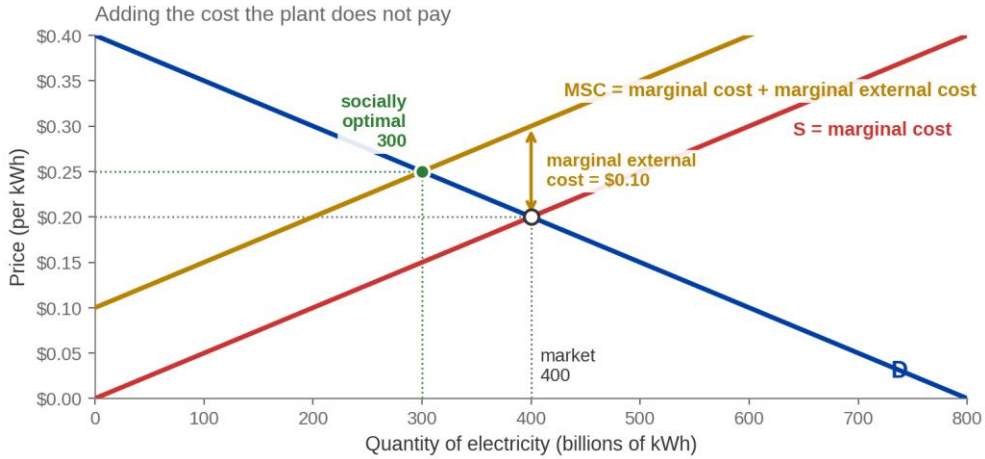
None of it appears on the supply curve, because the plant is not billed for any of it. The curve is an honest picture of the plant's costs. It is a dishonest picture of the electricity's costs.

NEGATIVE EXTERNALITY — a spillover COST borne by a bystander, and no price anywhere reflects it.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Adding the Cost Nobody Bills

MARGINAL SOCIAL COST = MARGINAL PRIVATE COST + MARGINAL EXTERNAL COST, and MEC = $0.10 throughout.

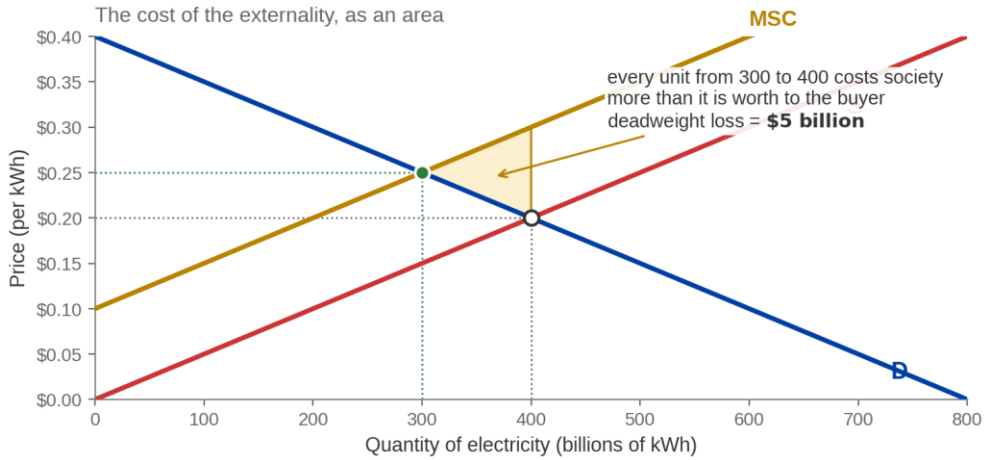


SOCIALLY OPTIMAL QUANTITY = 300 billion kWh, where demand meets MSC. Not 400, and not zero.

Redrawn from Exhibit 9.2.

Putting a Number on the Damage

Between 300 and 400, MARGINAL SOCIAL COST exceeds MARGINAL BENEFIT on every unit produced.

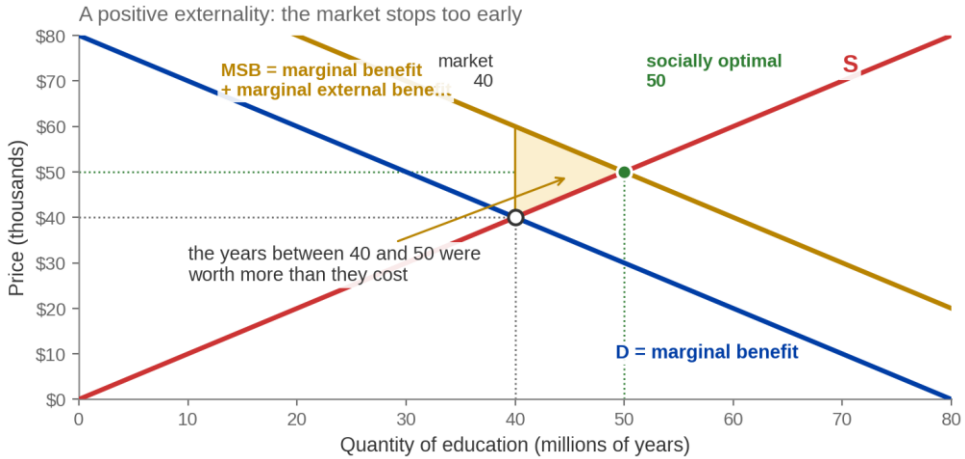


DEADWEIGHT LOSS = ½ × $0.10 × 100 = $5 billion — value that no one receives.

Redrawn from Exhibit 9.3.

Positive Spillovers

MARGINAL SOCIAL BENEFIT = MARGINAL PRIVATE BENEFIT + MARGINAL EXTERNAL BENEFIT, and MEB = $20 thousand.









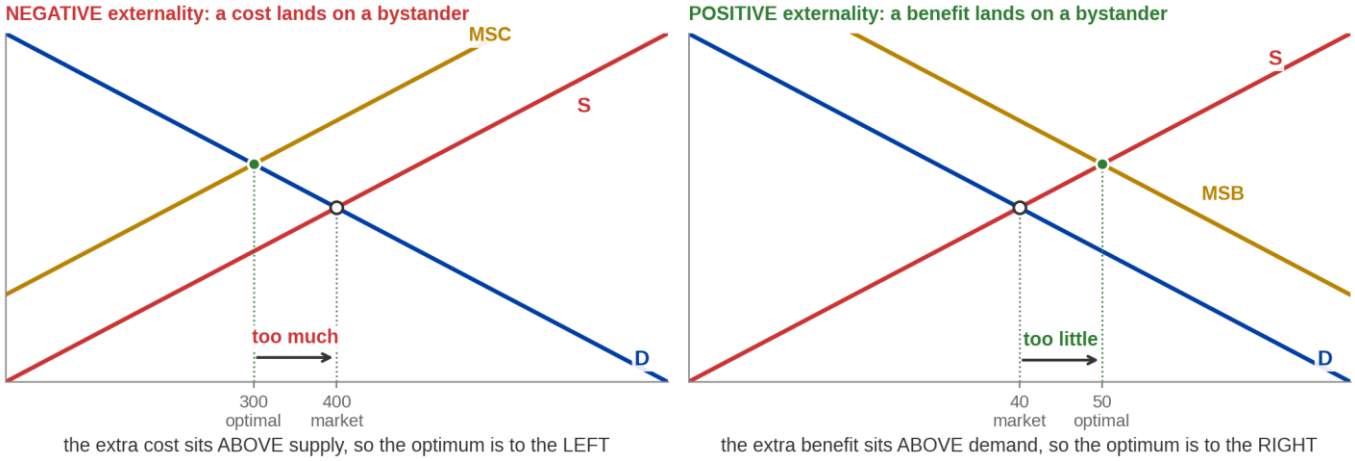


POSITIVE EXTERNALITY — a spillover BENEFIT. The market UNDER-produces: 40, not the optimal 50.

Redrawn from Exhibits 9.4 and 9.5.

Two Failures, One Diagnosis

Both markets clear, and both clear at the wrong quantity — the EXTERNALITY decides the direction.



the extra cost sits ABOVE supply, so the optimum is to the LEFT

the extra benefit sits ABOVE demand, so the optimum is to the RIGHT

NEGATIVE EXTERNALITY → OVER-PRODUCTION. POSITIVE EXTERNALITY → UNDER-PRODUCTION.

Pecuniary Externality

PECUNIARY EXTERNALITY — when a market exchange affects OTHER people through market prices.

A PECUNIARY EXTERNALITY is a TRANSFER through prices — not a MARKET FAILURE, and it creates no DEADWEIGHT LOSS.



The Silicon Valley Effect

NOT AN EXTERNALITY

Your company moved to San Jose and local housing demand rises and you pay more. You are IN this market. The higher price is the signal doing its job.

AN EXTERNALITY

The original residents there also need to pay more. They are not involved in your company's actions.

The test is not "was I made worse off from the price?" It is "was I a party to the transaction?"

PART 2

Can They Just Sort It Out Between Themselves?

Fred's Filter: A Bargain Worth Making

Fred's sweater plant discharges upstream. Anne's brewery draws the water downstream.
Profits per day:

	Without the filter	With the filter	Change
Fred (sweater plant)	$130	$100	−$30
Anne (brewery)	$90	$140	+$50
Both together	$220	$240	+$20

The filter is worth $50 to Anne and costs Fred $30, so any payment strictly between them works.
At $35: Fred ends at $135 (+$5), Anne at $105 (+$15). The $20 gain is split, not created twice.

PRIVATE BARGAINING: the filter creates the $20 SOCIAL GAIN. The payment only splits it — $5 to Fred, $15 to Anne.

Coase: Change Who Is Liable, and Watch What Does Not Change

COASE THEOREM — with clear PROPERTY RIGHTS and low TRANSACTION COSTS, private bargaining reaches an EFFICIENT allocation, whoever holds the right.

Cattle can stray onto a wheat farm. A fence costs $25,000.	If the RANCHER is liable	If the FARMER bears it
Damage is $30,000	Rancher builds the fence — $25,000	Farmer builds the fence — $25,000
Damage is $20,000	Rancher pays $20,000 in damages	Farmer absorbs $20,000



Ronald Coase

Read DOWN the columns: they are identical. The law decides who pays. It does not decide what gets done.

The EFFICIENT OUTCOME is invariant to the assignment of PROPERTY RIGHTS. Only the DISTRIBUTION changes.

Why We Cannot Just Leave It to Bargaining

Three conditions on the COASE THEOREM — and real externalities break all three.

TRANSACTION COSTS

Negotiating takes lawyers, time and information. If that costs more than the $20 on the table, the efficient deal does not get struck.



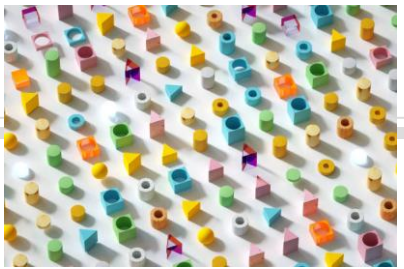
UNCLEAR PROPERTY RIGHTS

Bargaining needs somebody to hold the right. Who owns the air above a city, or an aquifer, or the ocean? Often the law does not say.



TOO MANY PARTIES

Fred and Anne is two people. A power plant and 100,000 downwind households cannot get into a room, and each has a reason to stay out.



TRANSACTION COSTS, ill-defined PROPERTY RIGHTS and FREE RIDING are why private bargaining usually fails.

PART 3

What Government Can Do About It

Two Ways In

INTERNALIZING THE EXTERNALITY — making the decision-maker face the full social cost.

COMMAND AND CONTROL



Government regulates the allocation directly. Produce 300, not 400. Fit this scrubber. Cap injection at 8,000 barrels a day. Simple, fast and effective — but it fixes the technology in place, because nothing rewards a firm for beating the standard.

MARKET-BASED

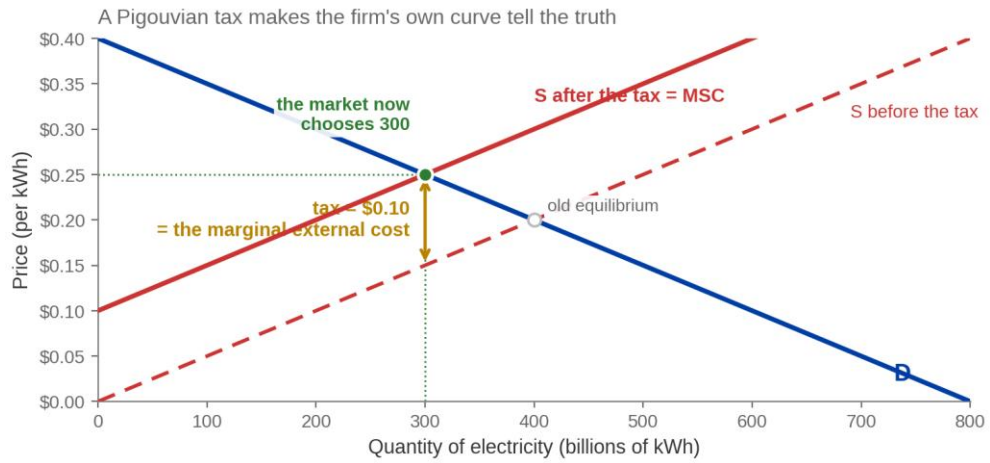


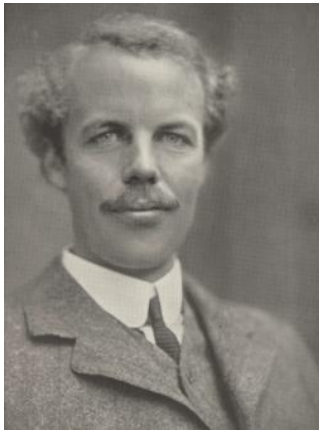
Government changes the prices and lets firms choose: a tax per unit of harm, a subsidy per unit of benefit, a tradable permit. Slower to design — but it puts the abatement where it is cheapest, because the firm that can cut cheaply now has a reason to cut more.

COMMAND-AND-CONTROL sets the QUANTITY. MARKET-BASED POLICY sets the PRICE and lets firms choose the quantity.

A Tax That Is Exactly the Right Size

PIGOUVIAN TAX = MARGINAL EXTERNAL COST = $0.10 per kWh, which lifts supply onto MSC.



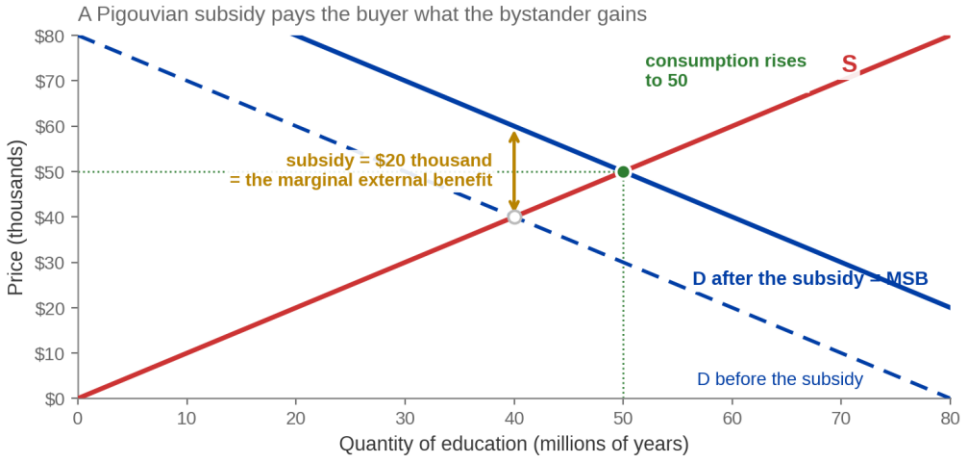


Arthur Pigou

TAX INCIDENCE: buyers pay $0.25, sellers keep $0.15. Revenue of $30 billion is a TRANSFER, not the efficiency gain.

And a Subsidy That Is Exactly the Right Size

PIGOUVIAN SUBSIDY = MARGINAL EXTERNAL BENEFIT = $20 thousand, which lifts demand onto MSB.



Quantity rises from 40 to the SOCIALLY OPTIMAL 50, and the DEADWEIGHT LOSS is eliminated.

Selling the Right to Pollute

CAP AND TRADE — government fixes the total QUANTITY of emissions and issues permits for it, then lets firms buy and sell them.

WHO SELLS

A firm that can cut emissions cheaply cuts more than it has to and sells the permits it no longer needs. The cut costs it less than the permit fetches.

WHO BUYS

A firm facing high abatement costs buys permits instead. Paying for the right is cheaper than retooling.

WHAT IT CREATES

A property right in something that never had one. The cap fixes the total; the market decides who does the cutting, and it picks the cheapest cutters.

A permit price equalizes MARGINAL ABATEMENT COST across firms — least-cost for any given CAP.

You Are a Producer of Externalities Too



Should you be charged by the bag?

Under a flat annual fee for rubbish collection, your marginal cost of throwing one more bag away is zero. So the landfill fills faster than anyone would choose.

Pay-as-you-throw charges by volume instead. The cost of the next bag stops being zero, and households throw less and recycle more — the same mechanism as the tax on the plant, pointed at a consumer.

A flat fee sets your MARGINAL COST of the next bag to zero. Consumers create NEGATIVE EXTERNALITIES too.

Image: Acemoglu, Laibson & List, Microeconomics 3e

PART 4

Goods the Market Will Not Supply

Two Characteristics of a Good

Ask them in this order. The answers are independent, which is why there are four kinds of good and not two.

IS IT RIVAL?



If I consume it, can you still consume it? RIVAL — I eat the apple, you cannot; one trout, one person. NONRIVAL — we both stream the same film. The army defending me defends you at no extra cost.

IS IT EXCLUDABLE?



Can somebody who has not paid be kept out? EXCLUDABLE — no payment, no apple; no subscription, no stream. NONEXCLUDABLE — the army cannot defend the houses that paid and skip the one that did not.

RIVALRY is a fact about the good. EXCLUDABILITY is a fact about whether you can charge for it.

Four Kinds of Good

PRIVATE GOODS · CLUB GOODS · COMMON POOL RESOURCES · PUBLIC GOODS

	EXCLUDABLE — market can charge	NONEXCLUDABLE — market cannot charge
RIVAL **can be used up**	PRIVATE GOODS clothes, food, furniture, a hamburger, a car The market supplies these well. 	COMMON POOL RESOURCES ocean fish, groundwater, natural forest, a congested road Overused — all of Part 5. 
NONRIVAL **shared freely**	CLUB GOODS streaming, satellite TV, campus Wi-Fi, a country club Supplied privately, rarely competitively. 	PUBLIC GOODS national defense, an early-warning system mosquito abatement Barely supplied at all. 

The market supplies EXCLUDABLE goods. NONEXCLUDABILITY is what defeats it — and it defeats it twice over.

Why Nobody Will Sell You National Defence

FREE-RIDER PROBLEM — an individual does not pay for a good because it is NONEXCLUDABLE, and consumes it anyway.

NOBODY KNOWS THEIR SHARE



Ask people to post a cheque to Washington for national defence. Even the willing cannot work out what their share is — the good is nonrival, so there is no quantity to divide.

NOBODY CAN BE MADE TO



If you send nothing, will the army leave your street undefended? It cannot. Not paying costs you nothing at all.

SO IT IS TAXED

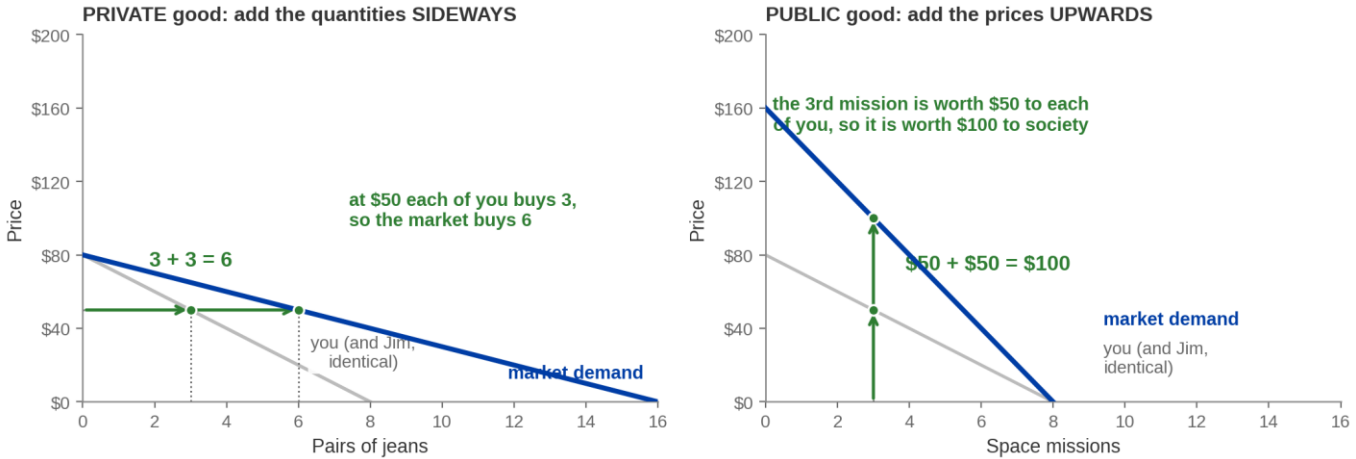


Government supplies the good and makes paying compulsory. That is what taxation is for — not a moral claim, a solution to this exact problem.

FREE RIDING is individually rational and collectively ruinous — hence COMPULSORY TAXATION.

Adding Up Demand — Sideways, Then Upwards

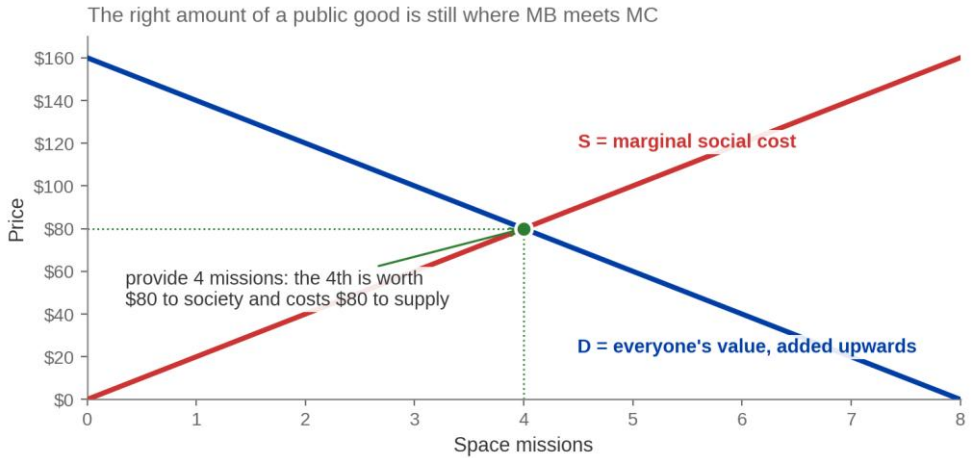
Private good and Public good build different market demand curves. Same Public good quantities are consumed by all individuals.



PRIVATE GOOD — sum QUANTITIES horizontally. PUBLIC GOOD — sum PRICES vertically.

How Much Should We Provide?

Provide until MARGINAL SOCIAL BENEFIT = MARGINAL SOCIAL COST — the same rule as ever.

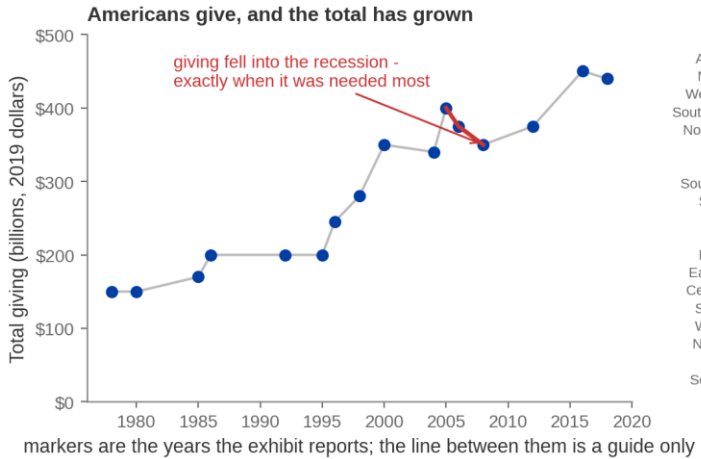


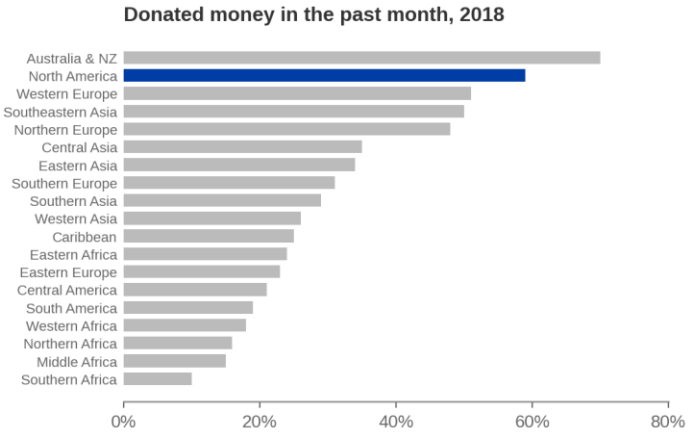
Q* = 4 missions at $80, where MSB = MSC. Only the DEMAND curve was built differently.

Redrawn from Exhibit 9.13.

Some of It Does Get Paid For Voluntarily

PRIVATE PROVISION OF A PUBLIC GOOD — voluntary contributions toward a good nobody can be charged for.





Hundreds of billions a year — and PROCYCLICAL: it fell into the 2008 recession, when the need was greatest.

What Charity Cannot Fix





Which of these would you rather save?

Most people say the panda. Ochre sea stars are a keystone species — remove them and the whole intertidal community collapses. Pandas are charismatic. Sea stars are load-bearing.

Voluntary giving allocates by what moves donors, which is not the same as allocating by where the need is greatest. Add the two problems from the last slide — the total is variable, and it falls in downturns — and private provision cannot be the whole answer.

FREE RIDING under-supplies the good; donor preference MISALLOCATES what is supplied. Two separate failures.

Image: Acemoglu, Laibson & List, Microeconomics 3e

PART 5

Goods the Market Uses Up

Rival, and Nobody Can Be Kept Out

Section 9.5 · the externality with nobody to bargain with



Why would people who fish for a living vote for limits on their own catch?

A common pool resource is rival — the fish you land is not there for anyone else — and nonexcludable, so nobody can be stopped from fishing.

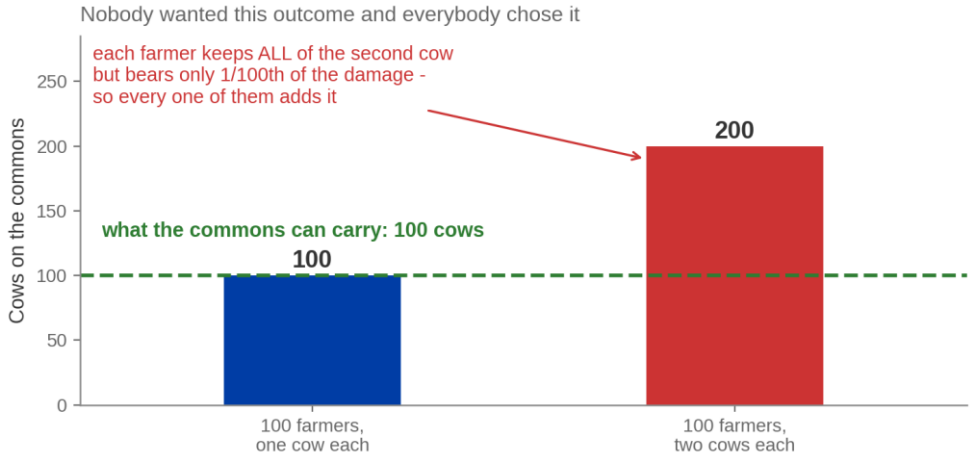
That combination means every boat's catch imposes a cost on every other boat, and no boat is billed for it. It is the negative externality from Part 1, with the property right missing.

TRAGEDY OF THE COMMONS — OPEN ACCESS plus DEPLETION THROUGH USE. A negative externality with no owner to bargain with.

Image: Acemoglu, Laibson & List, Microeconomics 3e

A Hundred Farmers, a Hundred Cows, and One Extra Each

The commons carries 100 cows. One hundred farmers with one cow each is exactly sustainable.



Each farmer captures the whole MARGINAL PRIVATE BENEFIT and bears one hundredth of the MARGINAL SOCIAL COST.

Why Cattle Are Not Extinct and Bison Nearly Were

Same animal, same uses, opposite outcomes — and the only difference was who owned them.



Owned privately: maintain, protect, conserve, trade.
Held in common: neglect it, and take yours first.

PRIVATE OWNERSHIP

Assign the right and the owner bears the whole cost of depletion. Ranching, fishing quotas, land titles.

REGULATION

Cap the take. Catch limits, closed seasons, size rules — with enforcement and real penalties.

A CHARGE FOR USE

Price the access. A congestion charge, a grazing fee — Part 3's tax, pointed at a resource.

PROPERTY RIGHTS, REGULATION or a PIGOUVIAN CHARGE — all three INTERNALIZE the externality.

Chapter 9: The Full Picture

	What it says	Key terms
Externalities	A cost or benefit lands on somebody outside the deal, so the price is wrong.	externality, MSC, MSB, deadweight loss
Which way it fails	Negative → the market makes too much. Positive → too little.	negative and positive externality
Private fixes	Bargaining works if rights are clear, costs are low and parties are few.	Coase theorem, transaction costs
Government fixes	Mandate the quantity, or price the harm and let firms choose.	command-and-control, Pigouvian tax and subsidy
Public goods	Nonrival and nonexcludable, so nobody can be charged. Add demand upwards.	free rider, nonrival, nonexcludable
Common pool	Rival and nonexcludable, so it gets used up. Assign, regulate or charge.	common pool resource, tragedy of the commons

Every row is one sentence: when PRIVATE cost and benefit differ from SOCIAL cost and benefit, the price misinforms.

KEY TAKEAWAYS

Chapter 9: Externalities and Public Goods

- EXTERNALITY — a cost or benefit falling on somebody who is not party to the transaction. The market still clears, at the wrong quantity.
- NEGATIVE EXTERNALITY → over-production; POSITIVE EXTERNALITY → under-production. The extra curve is always above the old one; what differs is which curve.
- COASE THEOREM — with clear PROPERTY RIGHTS and low TRANSACTION COSTS, the efficient action does not depend on who is liable. Only the distribution does.
- PIGOUVIAN TAX = MARGINAL EXTERNAL COST. It lifts the firm's private cost curve onto MARGINAL SOCIAL COST, so the firm picks the social optimum unprompted.
- NONEXCLUDABILITY is the deep problem: FREE RIDING when the good is NONRIVAL, TRAGEDY OF THE COMMONS when it is RIVAL.